Perfect squares

1 State whether the following numbers are perfect squares:

a 6 b 25 c 49 d 44
e 18 f 144 g 36 h 12

2 Evaluate the following:

a 4^2 b 7^2 c 9^2 d 12^2
e 9^2+6 f 3^2+7^2 g 7^2+9^2 h 100^2+10^2
i 7^2-3^2 j 9^2-8^2 k 12^2-6^2 l 100^2-10^2

3 Harry was working out $2^2 \times 5^2$ and thought that he could simplify the expression using the fact that $2 \times 5 = 10$.

a Evaluate $2^2 \times 5^2$, by first expanding and evaluating each square.
b Now, using the fact $2 \times 5 = 10$, evaluate 10^2.
c Is $2^2 \times 5^2 = (2 \times 5)^2$ a true statement?
d Calculate $4^2 \times 2^2$ using the above method.

Square roots

4 Given that we know $5 \times 5 = 25$. What is $\sqrt{25}$?

5 Evaluate the following:

a $\sqrt{4}$ b $\sqrt{16}$ c $\sqrt{121}$ d $\sqrt{144}+\sqrt{64}$
e $\sqrt{144}-\sqrt{64}$ f $\sqrt{6^2+8^2}$ g $\sqrt{13^2-12^2}$ h $\sqrt{13^2}-\sqrt{12^2}$

Higher roots

6 Are the following statements true or false?

a $\sqrt[3]{1000}=10$ b $\sqrt[3]{64}=4$

7 Evaluate the following:

a $\sqrt[3]{27}$ b $\sqrt[3]{64}$ c $\sqrt[3]{1000}$ d $\sqrt[3]{343}$

8 Evaluate the following:

a $\sqrt[4]{16}$ b $\sqrt[4]{256}$ c $\sqrt[5]{100\,000}$ d $\sqrt[5]{32}$

9 Solve $x^3 = 64$.

10 Find the side length of a cube with a volume of $27\ \text{cm}^3$.

11 Evaluate the following:

a $\sqrt[3]{512} \times \sqrt[3]{64}$

b $\sqrt[3]{64} \div \sqrt[3]{8}$

c $\sqrt[3]{8} \times \sqrt[3]{1000}$

d $\sqrt[3]{64} \div \sqrt{16}$

e $6^2 \div \sqrt[3]{8}$

f $\sqrt[3]{1000} \div \sqrt{100}$

g $8^3 \div \sqrt[3]{64}$

h $\sqrt[3]{125} \times \sqrt[5]{32} \div \sqrt[3]{8}$